STVDIVM VRBIS

From centroidal to whole-body models for legged locomotion: a comparative analysis

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SUMMARY

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General Problem

Bipedal locomotion in humanoid robots is a complex challenge due to dynamic instability, limitations of ground reaction forces and the complexity of multi-body systems.

In robotics, **Model Predictive Control** (MPC) is a common approach to plan and stabilise locomotion, but the choice of dynamic model directly affects performance, stability and computational cost.

We will analyse and compare three modelling approaches

Full-Body model

Centroidal model

Kino-Dynamic model

General MPC

Performance of MPC models depends on the choice of dynamic model and the length of the horizon.

$$\min_{\underline{x}\,\underline{u}} \sum_{t=0}^{T-1} l_t(x_t, u_t) + l_T(x_T)$$

$$\begin{aligned}
s.t. \qquad & x_0 = \hat{x}_0 \\
& 0 = h_T(x_T) \\
\forall t = 0..T-1, \quad & x_{t+1} = f(x_t, u_t) \\
& 0 = h_i(x_i, u_i) \\
& 0 \leq g_i(x_i, u_i)
\end{aligned}$$

General MPC

Performance of MPC models depends on the choice of dynamic model and the length of the horizon.

$$\min_{\underline{x}\,\underline{u}} \sum_{t=0}^{T-1} \underbrace{l_t(x_t, u_t)}_{1} + \underbrace{l_T(x_T)}_{2}$$

$$\begin{aligned} s.t. \quad & x_0 = \hat{x}_0 \\ & 0 = h_T(x_T) \\ \forall t = 0..T-1, \quad & x_{t+1} = f(x_t, u_t) \\ & 0 = h_i(x_i, u_i) \\ & 0 \leq g_i(x_i, u_i) \end{aligned}$$

Running cost

1

Terminal cost

2

General MPC

Performance of MPC models depends on the choice of dynamic model and the length of the horizon.

$$\min_{\underline{x}\,\underline{u}} \sum_{t=0}^{T-1} l_t(x_t, u_t) + l_T(x_T)$$

$$s.t. \quad \underline{x_0 = \hat{x}_0}$$

$$0 = h_T(x_T)$$

$$\forall t = 0..T-1, \quad \underline{x_{t+1} = f(x_t, u_t)}$$

$$0 = h_i(x_i, u_i)$$

$$0 \leq g_i(x_i, u_i)$$

Initial measured state of the system

3

Discretized dynamics

4

General MPC

Performance of MPC models depends on the choice of dynamic model and the length of the horizon.

$$\min_{\underline{x}\,\underline{u}} \sum_{t=0}^{T-1} l_t(x_t, u_t) + l_T(x_T)$$

$$\begin{aligned} s.t. \quad & x_0 = \hat{x}_0 \\ & 0 = h_T(x_T) \\ \forall t = 0..T-1, \quad & x_{t+1} = f(x_t, u_t) \\ & 0 = h_i(x_i, u_i) \\ & 0 \leq g_i(x_i, u_i) \end{aligned}$$

5

Equality constraint action on the terminal state

Equality constraint

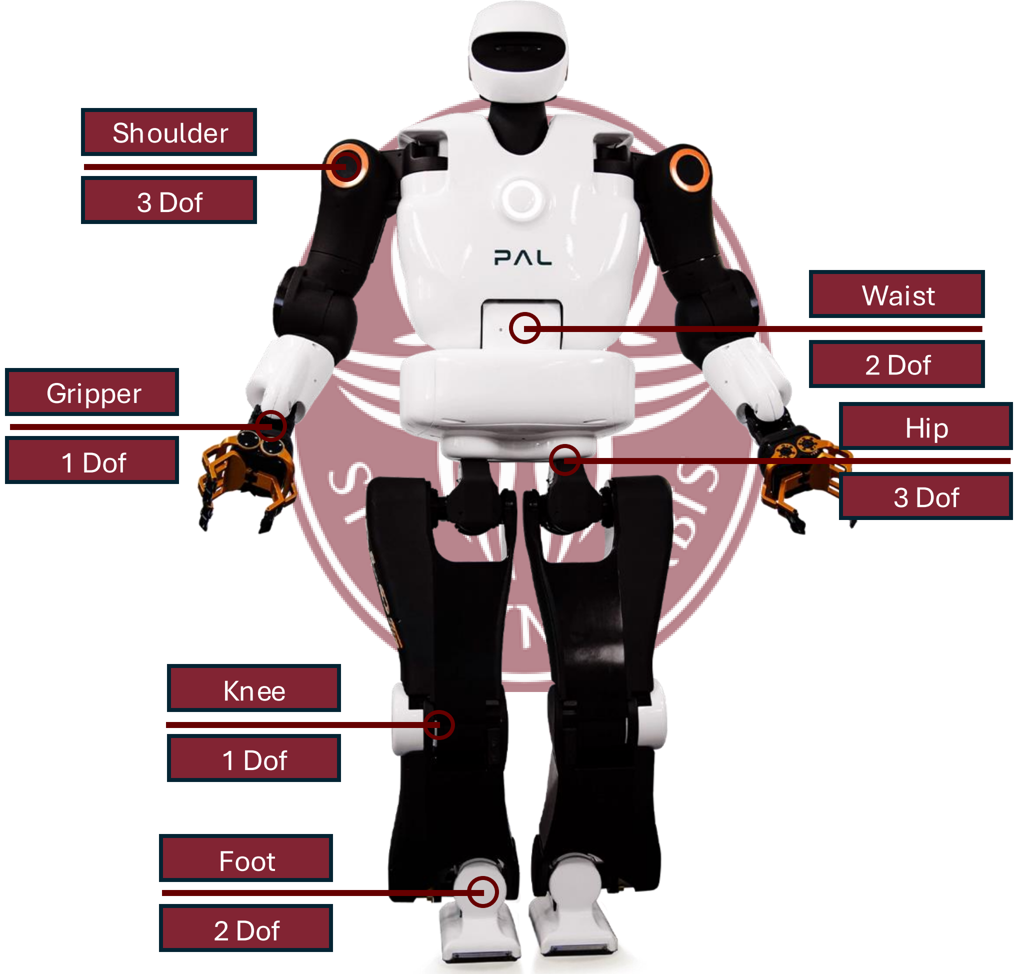
6

7

Inequality contraints

Talos Robot Model

The Robot used in simulation is
TALOS (PAL Robotics 2017)



Actuated joints $n_j = 22$

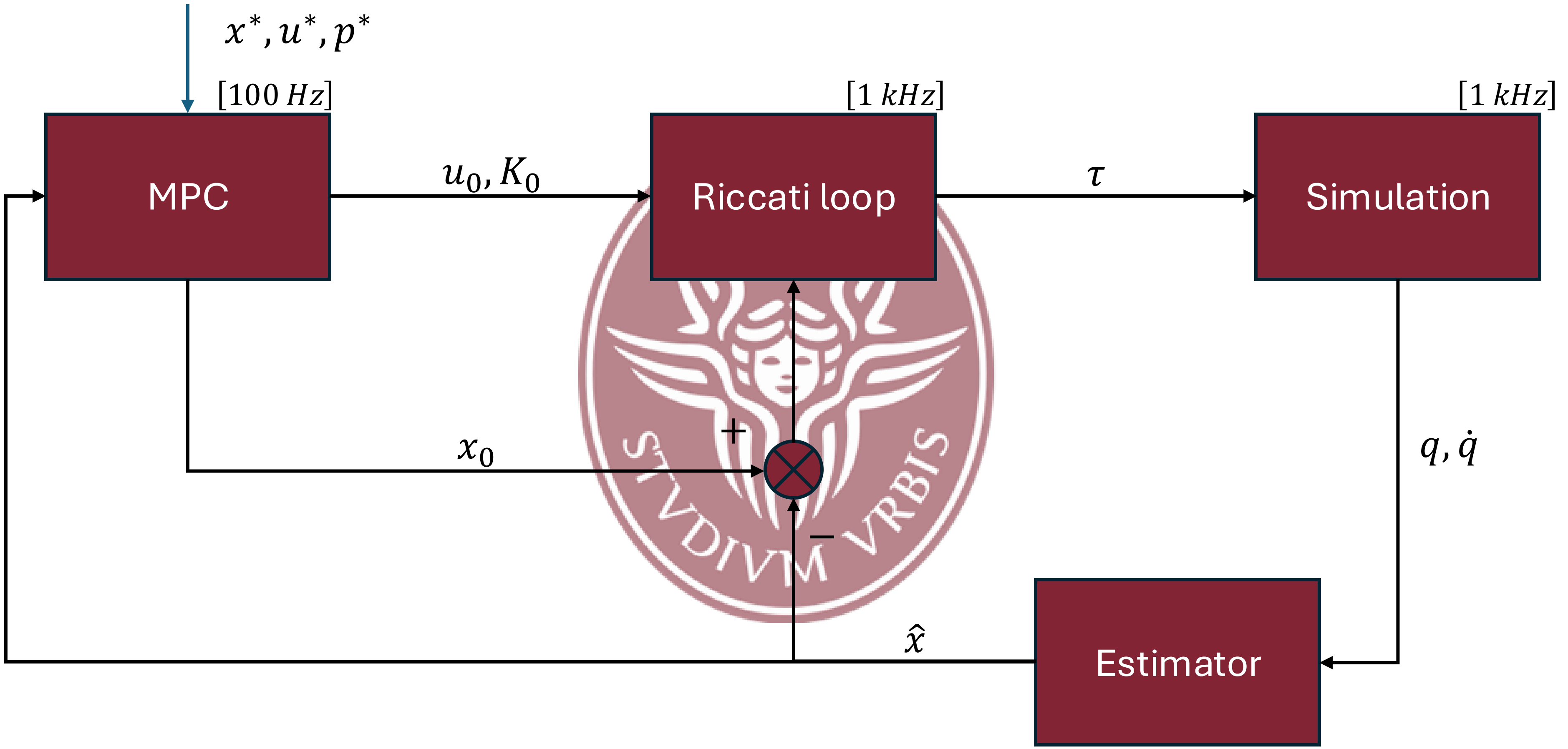
12 legs' joints

2 torso's joints

8 arms' joints

FULL-BODY MPC

Block Scheme



$x = (q, \dot{q})$

$u = (\tau_1, \dots, \tau_{n_j})$

State and Input

This model provides a complete description of the multi-contact, whole body dynamics

While being computationally expensive, the MPC loop returns an optimal torque that could be used directly for the whole-body control (but only at 100Hz!)

State $x = (q, \dot{q}) \in \mathbb{R}^{n_q + n_v}$

$$q = \begin{bmatrix} p_b \\ r_b \\ q_a \end{bmatrix}$$

FLOATING BASE

$$p_b = [p_x, p_y, p_z]^T$$
$$r_b = [a, b, c, d]^T$$

ACTUATED JOINTS

$q_a \in \mathbb{R}^{n_j}$

For TALOS 22 actuated joints

$$\dot{q} = \begin{bmatrix} \dot{p}_b \\ \omega_b \\ \dot{q}_a \end{bmatrix}$$

F. BASE VELOCITY

$$\dot{p}_b = [\dot{p}_x, \dot{p}_y, \dot{p}_z]^T$$
$$\omega_b = [\omega_x, \omega_y, \omega_z]^T$$

Therefore, in our scenario:
$q \in \mathbb{R}^{29}$
$\dot{q} \in \mathbb{R}^{28}$

Control

$$u = (\tau_1, \dots, \tau_{n_j}) \in \mathbb{R}^{n_j}$$

With this in mind, we can formulate our MPC problem

Dynamic Model (1)

Karush-Kuhn-Tucker (KKT) conditions

$$\begin{bmatrix} M(q) & J_c(q)^T \\ J_c(q) & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ -\lambda \end{bmatrix} = \begin{bmatrix} S^T\tau - b(q,\dot{q}) \\ -\dot{J}_c(q)\dot{q} \end{bmatrix}$$

- $M \in \mathbb{R}^{n_v \times n_v}$ inertial matrix
- $b \in \mathbb{R}^{6K \times n_v}$ generalized non linear forces
- $\lambda \in \mathbb{R}^{6K}$ contatenated contact wrenches
- $J_c \in \mathbb{R}^{6K \times n_v}$ contact Jacobians
- $\tau \in \mathbb{R}^{n_j}$ vector of commanded joint torque

- $S = \begin{bmatrix} 0 & I_{n_j} \end{bmatrix} \in \mathbb{R}^{n_j \times n_v}$ actuation matrix

In order to use the KKT conditions as our dynamics, we need to express the contat wrench stack $\boldsymbol{\lambda}$ as a function of our decision variables $(\boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{\tau})$

UNDERACTUATED ROBOT:
The floating base is not directly affected by commanded torques

Dynamic Model (2)

By inverting the **Operational Space Inertia Matrix (OSIM)**

$$\Lambda^{-1} = J_c(q)M(q)J_c^T$$

using a **special Cholewski decomposition**, we can always (even in the singular configurations) Express the wrench as a function of the decision variables

$$\lambda = \Phi(q, \dot{q}, \tau)$$

We can finally define our state space variables:

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix} \equiv \begin{bmatrix} x_1 \\ x_2 \end{bmatrix};\ u = \tau$$

KKT

STATE SPACE REPRESENTATION
expressed as a function of the decision variables

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -b(x_1, x_2) - S^T u + J_c(x_1)^T \Phi(\mathrm{x}, \mathrm{u})$$

Discretization

$$x_{t+1} = f(x_t, u_t)$$

4

Cost Function

The running cost aims to regularize the state, the input, the contact wrenches and the feet trajectory

$$l_t(x,u) = \|x - x^*\|^2_{D_x} + \|u - u^*(t)\|^2_{D_u} + \|p(x) - p^*(t)\|^2_{D_v} + \|\lambda(x,u) - \lambda^*(t)\|^2_{D_\lambda}$$

1

$$l_T(x) = \|x - x^*\|^2_{D_x}$$

2

with D_* being properly tuned diagonal weight matrices

x^* is the static half sitting configuration reference:
$x^* = [\mathbf{q}_{\mathbf{hs}}, \mathbf{0}_{\mathbf{nv}}]^{\mathrm{T}}$

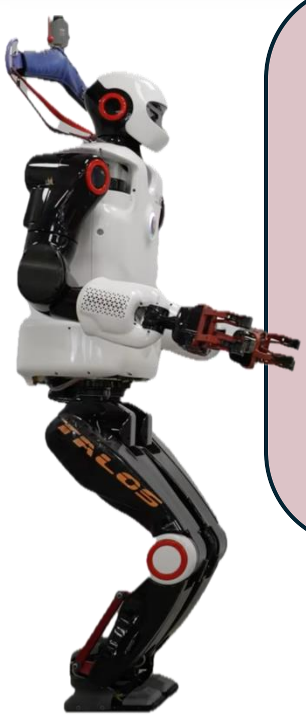
$u^*(t)$ is the gravity compensating torque given the current q:
$u^*(t) = \tau_{grav-comp}(q(t))$

$p(x), p^*(t)$ are the current and desired foot placement (position + orientation)

$\lambda(x,u), \lambda^*(t)$ are the current and desired contact wrenches

Gait Generation

Define G = mg (gravity compensation force)



The contact wrench references λ^* handle the weight distribution of the robot on the two feet. Linear interpolations are used to move from a DOUBLE contact to a SINGLE contact configuration (and vice-versa)

Parametrize Spline Foot Trajectory

swing apex

~ zero velocity At take-off/landing

$$\lambda_l^* = \begin{bmatrix} f_l^* \\ \tau_l^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ G/2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_l^* = \begin{bmatrix} f_l^* \\ \tau_l^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

PHASE 0

Half-sitting Configuration (DOUBLE CONTACT)

$$\lambda_r^* = \begin{bmatrix} f_r^* \\ \tau_r^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ G/2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

PHASE 1

Weight Shifting (SINGLE CONTACT)

$$\lambda_r^* = \begin{bmatrix} f_r^* \\ \tau_r^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ G \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

PHASE 2

Step execution

Inequality Constraints

Joint and Torque Limits

$$g_i^l(x_i, u_i) = \begin{bmatrix} X^j x_i - q_l \\ q_u - X^j x_i \\ \tau_u - u_i \\ \tau_u + u_i \end{bmatrix} \geq 0$$

7

- $q_l, q_u \in \mathbb{R}^{n_j}$ lower and upper joint limits
- $\tau_u \in \mathbb{R}^{n_j}$ torque limits
- $X^j \in \mathbb{R}^{n_j \times n_x}$ selection matrix for actuated joints

Contact constraints

$$g_{i,k}^{\lambda}(x_i, u_i) = A\lambda_k(x_i, u_i) \geq 0$$

$A \in \mathbb{R}^{17\times 6}$ whose elements depend on friction coefficient and feet parameters

$$|\boldsymbol{f}_x| \leq \boldsymbol{\mu} \boldsymbol{f}_z$$
$$|\boldsymbol{f}_y| \leq \boldsymbol{\mu} \boldsymbol{f}_z$$
$$\boldsymbol{f}_z > \mathbf{0}$$

Equality Constraints

DCM capturability condition

$$h_T(x_T) = c_T + \alpha \dot{c}_T - d^*$$

5

- $c_T, \dot{c}_T$ position and velocity of the CoM at the end of the prediction horizon
- α DCM time-constant
- d^* desired DCM position at the end of the horizon (center of the support polygon)

We can now solve the MPC problem and compute the optimal control and state sequence ($\underline{x}$, $\underline{u}$) using an augmented Lagrangian DDP algorithm.

Riccati Loop

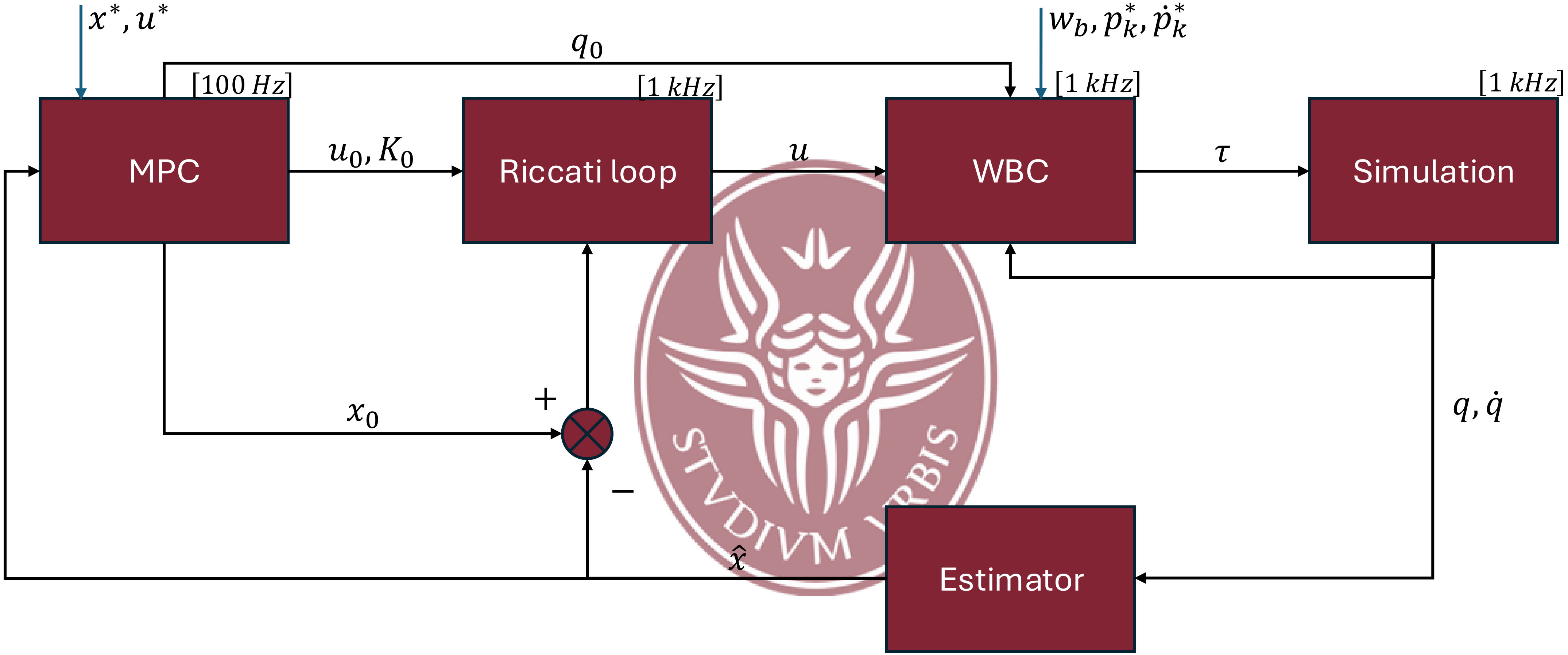
The MPC Loop is too slow to perform optimal real-time control

The Riccati Loop runs at 1kHz and “corrects” the reference input given by the MPC Loop

$$u = \tau = u_0 + K_0(x_0 \ominus \hat{x})$$

CENTROIDAL MPC

Block Scheme



$x = (c, h_l, h_a)$

$u = (\lambda_1, \dots, \lambda_K)$

State and Controller

The State is
$x = (c, h_l, h_a)$

$c \in \mathbb{R}^3$ is the CoM

$h_l = m\dot{c} \in \mathbb{R}^3$ is the linear momentum

$h_a \in \mathbb{R}^3$ is the angular momentum

$h_g = (h_l, h_a) \in \mathbb{R}^6$ is the total centroidal momentum of the system

The Controller is
$u = (\lambda_1 \dots \lambda_k) \in \mathbb{R}^{6K}$

$\lambda_k = (f_k, \tau_k)$ is the wrench produced by contact k

Dynamic Model

Nonlinear dynamics of the reduced model

$$\dot{h}_l = \sum_{k=1}^{K} f_k + mg$$

$f_k \in \mathbb{R}^3$ linear part of the corresponding ground reaction forces

$g \in \mathbb{R}^3$ gravitational component

$$\dot{h}_a = \sum_{k=1}^{K} (r_k^* - c) \times f_k + \tau_k$$

$\tau_k \in \mathbb{R}^3$ angular part of the corresponding ground reaction forces

$r_k^* \in \mathbb{R}^3$ translation of contact k in world frame

Cost Function

The cost formulation of the centroidal scheme aims to regularize the CoM trajectory of the system and prevent unstable behaviours.

It cannot provide any way to act on the multi-body configuration of the robot, therefore the trajectories calculated by the MPC loop may not feasible and may not guarantee tracking of the end effector references

$$l_t(x,u) = \|x - x^*\|_{D_x}^2 + \|u - u^*(t)\|_{D_u}^2 + \left\|\dot{h}_g\right\|_{D_h}^2$$

$$l_T(x) = \|x - x^*\|_{D_x}^2$$

Constraints

$$g^{\lambda}_{i,k}(x_i, u_{i,k}) = A\lambda_k(x_i, u_{i,k}) \geq 0$$

Contact wrench inequality

$$h_T(x_T) = c_T + \alpha\dot{c}_T - d^*$$

DCM capturability condition on the terminal node

Torque Computation (1)

Step 1. Riccati feedback law

$$u = \lambda = u_0 + K_0(x_0 - \hat{x}(q, \dot{q}))$$

$\hat{x}(q, \dot{q})$ is the centroidal state

Step 2. Inverse dynamics QP to evaluate the final joint torques

Torque Computation (2)

$$\min_{\tau,\delta\ddot{q},\delta\lambda} \|\delta\ddot{q}\|^2_{W_q} + \|\delta\lambda\|^2_{W_\lambda} + \left\|\dot{h}_g - \dot{h}_g^*\right\|^2_{W_h} + \|\ddot{w}_b - \ddot{w}_b^*\|^2_{W_r} + \sum_{k=1}^{K}\|\ddot{p}_k - \ddot{p}_k^*\|^2_{W_p}$$

s.t

$$M(\ddot{q}^* + \delta\ddot{q}) + b = S^T\tau + \mathrm{J}_\mathrm{c}^\mathrm{T}(\lambda + \delta\lambda)$$

$$J_c(\ddot{q}^* + \delta\ddot{q}) + \dot{J}_c\dot{q} = 0$$

$$\lambda + \delta\lambda \in \mathcal{K}$$

$$-\tau_u \leq \tau \leq \tau_u$$

Torque Computation (3)

$$\min_{\tau,\delta\ddot{q},\delta\lambda} \|\delta\ddot{q}\|^2_{W_q} + \|\delta\lambda\|^2_{W_\lambda} + \left\|\dot{h}_g - \dot{h}_g^*\right\|^2_{W_h} + \|\ddot{w}_b - \ddot{w}_b^*\|^2_{W_r} + \sum_{k=1}^{K}\|\ddot{p}_k - \ddot{p}_k^*\|^2_{W_p}$$

s.t

$$M(\ddot{q}^* + \delta\ddot{q}) + b = S^T\tau + \mathrm{J}_c^{\mathrm{T}}(\lambda + \delta\lambda)$$

$$J_c(\ddot{q}^* + \delta\ddot{q}) + \dot{J}_c\dot{q} = 0$$

$$\lambda + \delta\lambda \in \mathcal{K}$$

$$-\tau_u \leq \tau \leq \tau_u$$

- $w_b \in SO(3)$ is the base frame rotation
- $\mathcal{K}$ is the wrench cone defined by the slippage condition

Torque Computation (4)

$$\min_{\tau,\delta\ddot{q},\delta\lambda} \|\delta\ddot{q}\|^2_{W_q} + \|\delta\lambda\|^2_{W_\lambda} + \left\|\dot{h}_g - \dot{h}^*_g\right\|^2_{W_h} + \|\ddot{w}_b - \ddot{w}^*_b\|^2_{W_r} + \textstyle\sum_{k=1}^{K}\|\ddot{p}_k - \ddot{p}^*_k\|^2_{W_p}$$

s.t

$$M(\ddot{q}^* + \delta\ddot{q}) + b = S^T\tau + \mathrm{J}_c^{\mathrm{T}}(\lambda + \delta\lambda)$$

$$J_c(\ddot{q}^* + \delta\ddot{q}) + \dot{J}_c\dot{q} = 0$$

$$\lambda + \delta\lambda \in \mathcal{K}$$

$$-\tau_u \leq \tau \leq \tau_u$$

The references for joint acceleration, base rotation and end-effector placements are given by:

$$\ddot{q}^* = K_p^q(q_0 \ominus q) + K_d^q\dot{q}$$

$$\ddot{w}_b^* = K_p^w(\log_3(w_b)) + K_d^w\dot{w}_b$$

$$\ddot{p}_k^* = K_p^p(p_k^* \ominus p_k) + K_d^p(\dot{p}_k^* \ominus \dot{p}_k)$$

Torque Computation (5)

$$\min_{\tau,\delta\ddot{q},\delta\lambda} \|\delta\ddot{q}\|^2_{W_q} + \|\delta\lambda\|^2_{W_\lambda} + \left\|\dot{h}_g - \dot{h}_g^*\right\|^2_{W_h} + \|\ddot{w}_b - \ddot{w}_b^*\|^2_{W_r} + \sum_{k=1}^{K}\|\ddot{p}_k - \ddot{p}_k^*\|^2_{W_p}$$

s.t

$$M(\ddot{q}^* + \delta\ddot{q}) + b = S^T\tau + \mathrm{J}_c^{\mathrm{T}}(\lambda + \delta\lambda)$$

$$J_c(\ddot{q}^* + \delta\ddot{q}) + \dot{J}_c\dot{q} = 0$$

$$\lambda + \delta\lambda \in \mathcal{K}$$

$$-\tau_u \leq \tau \leq \tau_u$$

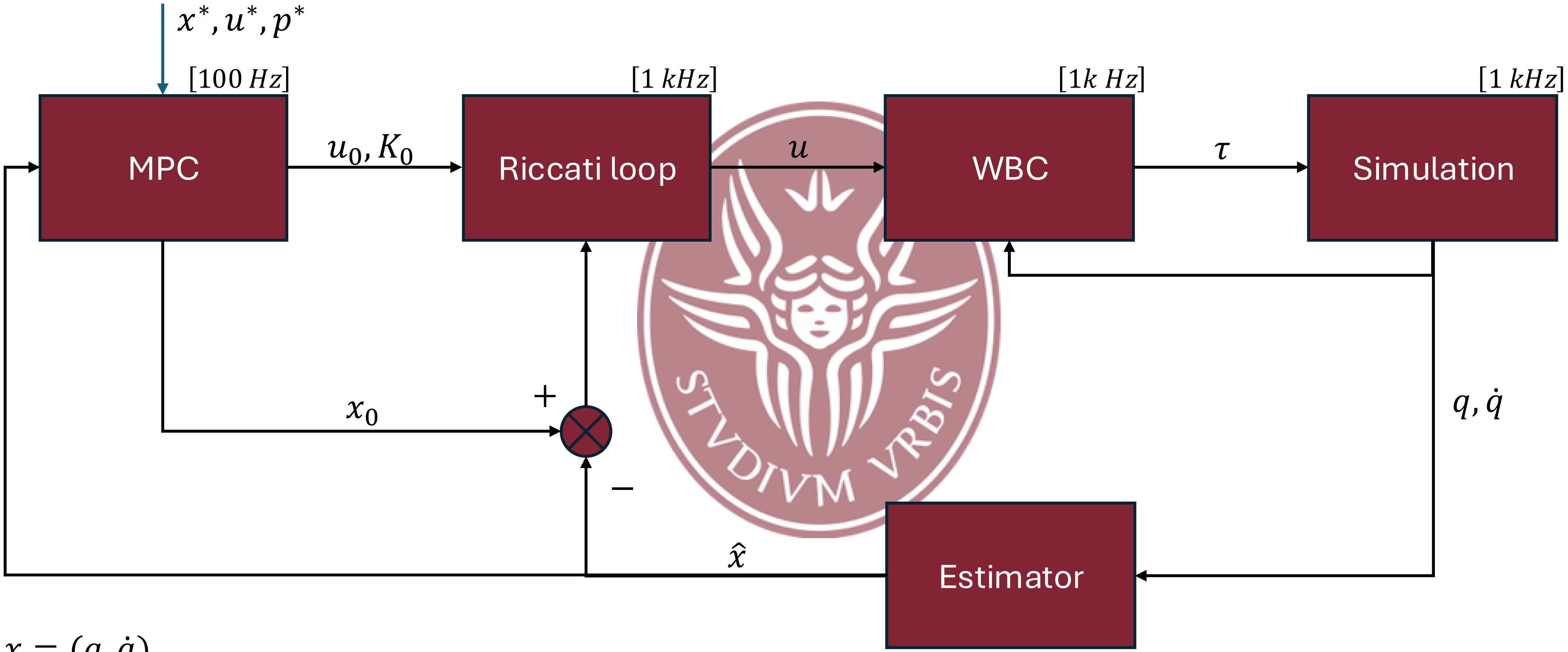
The accelerations associated to base rotation and feet placement, as well as the time derivative of momentum, are computed through:

$$\ddot{w}_b = J_b\ddot{q} + \dot{J}_b\dot{q}$$
$$\ddot{p}_k = J_k\ddot{q} + \dot{J}_k\dot{q}$$
$$\dot{h}_g = A_g\ddot{q} + \dot{A}_g\dot{q}$$

- $J_b \in \mathbb{R}^{3\times n_v}$: rotational part of the base Jacobian
- $J_k \in \mathbb{R}^{6\times n_v}$: Jacobian of the foot k
- A_g: centroidal momentum matrix

KINO-DYNAMICS MPC

Block Scheme



$x = (q, \dot{q})$

$u = (q_a, \lambda_1, \dots, \lambda_K)$

State and Controller

This model provides a compromise between the full dynamic model and the centroidal model

This model maps the dynamics of the CoM in the base frame of the robot

State: $x = (q, \dot{q}) \in \mathbb{R}^{n_q+n_v}$

Input: $u = (\ddot{q}_a, \lambda_1, \dots, \lambda_k) \in \mathbb{R}^{n_j+6K}$

Decomposition

We decompose the ceintroidal momentum into actuated and unactuated blocks

$$\dot{h}_g = [A_{g,u} \quad A_{g,a}] \begin{bmatrix} \ddot{q}_u \\ \ddot{q}_a \end{bmatrix} + \dot{A}_g \dot{q}$$

$$\ddot{q}_u = A_{g,u}^{-1}(\dot{h}_g - \dot{A}_g \dot{q} - A_{g,a}\, \ddot{q}_a)$$

$\ddot{q}_u \in se(3)$

$\ddot{q}_a \in \mathbb{R}^{n_j}$

$A_{g,u} \in \mathbb{R}^{6\times 6}$

$A_g \in \mathbb{R}^{6\times n}$ is Centroidal Momentum Matrix that maps the velocities of joints $\dot{q}$ in the ceintrodal momentum

Evaluation $\dot{h}_g$

$$\dot{h}_g = \begin{bmatrix} \sum_{k=1}^{K} f_k + mg \\ \sum_{k=1}^{K} \big(r_k(q) - c(q)\big) \times f_k + \tau_k \end{bmatrix}$$

Now the r_k and c are taken relative to the robot.

Cost Function (1)

$$l_t(x,u) = \|x - x^*\|^2_{D_x} + \|u - u^*(t)\|^2_{D_u} + \left\|\dot{h}_g\right\|^2_{D_h} + \|p(x) - p^*(t)\|^2_{D_p}$$

$\|x - x^*\|^2_{D_x}$	We want the robot to stay close to the half-sitting configuration
$\|u - u^*(t)\|^2_{D_u}$	Follow the contact sequence and acceleration of the CoM given by the user
$\left\|\dot{h}_g\right\|^2_{D_h}$	We want to minimize the total centroidal moment
$\|p(x) - p^*(t)\|^2_{D_p}$	Follow the foot trajectories given by the user

Cost Function (2)

$$l_T(x) = \|x - x^*\|^2_{D_x}$$

$\|x - x^*\|^2_{D_x}$

We want the robot at the final knot to be close to the half-sitting configuration.

Constraints (1)

Not having the conditions on contact accelerations applied naturally at the level of dynamics

We introduce a new constraint

$$h_i^v(x_i, u_i) = J_c\dot{q}$$

Constraints (2)

We also add the constraints already seen above

$$g^{\lambda}_{i,k}(x_i, u_{i,k}) = A\lambda_k(x_i, u_{i,k}) \geq 0$$

$$h_T(x_T) = c_T + \alpha \dot{c}_T - d^*$$

Implemented to ensure dynamic stability of centroidal motion

Constraits (3)

We also add the constraints already seen above

$$g^{\lambda}_{i,k}(x_i, u_{i,k}) = A\lambda_k(x_i, u_{i,k}) \geq 0$$

$$h_T(x_T) = c_T + \alpha \dot{c}_T - d^*$$

Implemented to ensure dynamic stability of centroidal motion

$$g^{l}_{i}(x_i, u_i) = \begin{bmatrix} X^j x_i - q_l \\ q_u - X^j x_i \end{bmatrix}$$

To account for the limitations of the joint

Inverse Dynamics QP (1)

We will define QP to calculate the torque τ that minimizes the acceleration $\delta\ddot{q}$ and wrench $\delta\lambda$ correction terms

$$\min_{\tau,\delta\ddot{q},\delta\lambda} \|\delta\ddot{q}\|^2_{W_q} + \|\delta\lambda\|^2_{W_\lambda}$$

$$s.t. \quad M(\ddot{q}^* + \delta\ddot{q}) + b = S^T\tau + \mathrm{J}_\mathrm{c}^\mathrm{T}(\lambda + \delta\lambda)$$

$$J_c(\ddot{q}^* + \delta\ddot{q}) + \dot{J}_c\dot{q} = 0$$

$$\lambda + \delta\lambda \in \mathcal{K}$$

$$-\tau_u \leq \tau \leq \tau_u$$

Inverse Dynamics QP (2)

We will define QP to calculate the torque τ that minimizes the acceleration $\delta\ddot{q}$ and wrench $\delta\lambda$ correction terms

$$\min_{\tau,\delta\ddot{q},\delta\lambda} \|\delta\ddot{q}\|^2_{W_q} + \|\delta\lambda\|^2_{W_\lambda}$$

$$s.t. \quad M(\ddot{q}^* + \delta\ddot{q}) + b = S^T\tau + \mathrm{J}_\mathrm{c}^\mathrm{T}(\lambda + \delta\lambda)$$

$$J_c(\ddot{q}^* + \delta\ddot{q}) + \dot{J}_c\dot{q} = 0$$

$$\lambda + \delta\lambda \in \mathcal{K}$$

$$-\tau_u \leq \tau \leq \tau_u$$

Dynamics model

Cone condition

Acceleration condition

Torque condition

Simulation (1)

Simulations have been performed using Talos robot, which has $n_j = 22$ joints $\Rightarrow$ dimensions of configuration space is $n_q = 29$ and the derivative $n_v = 28$

Prediction horizon is $T = 100$ with timestep $\delta t = 10\ ms$

Bullet simulation runs at 1 kHz. A MPC new policy is been evaluated every 10 simulation step

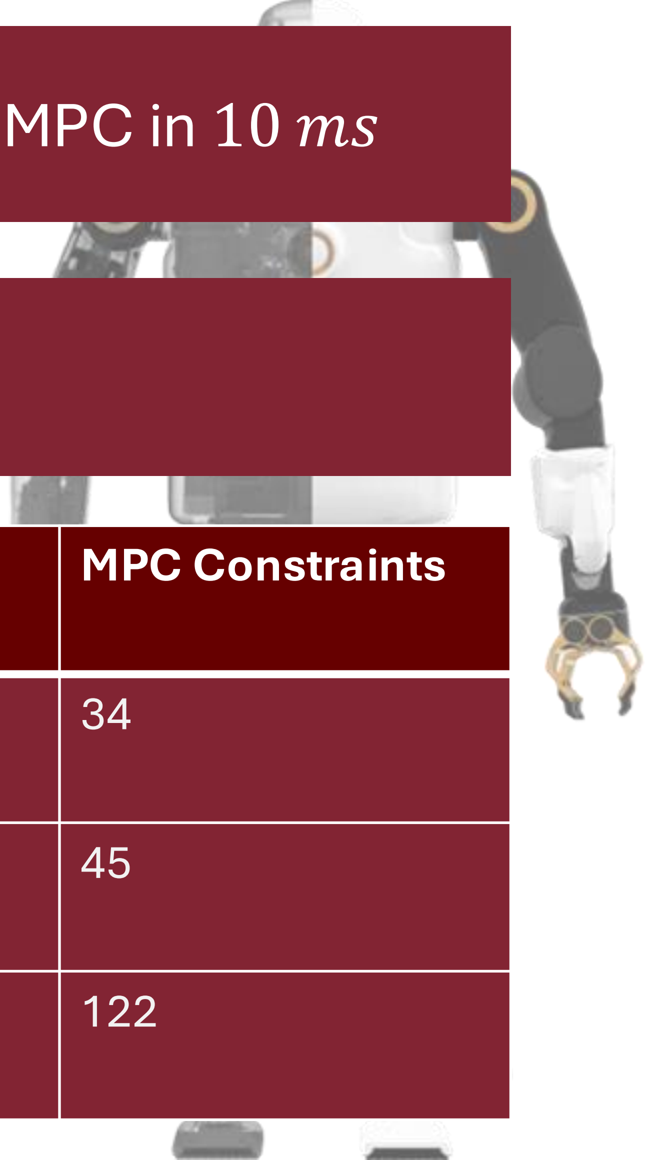
In simulations, attempts have been made to reduce the computational load by decreasing the prediction horizon or increasing the timestep but this leads to unstable behaviour

Simulation (2)

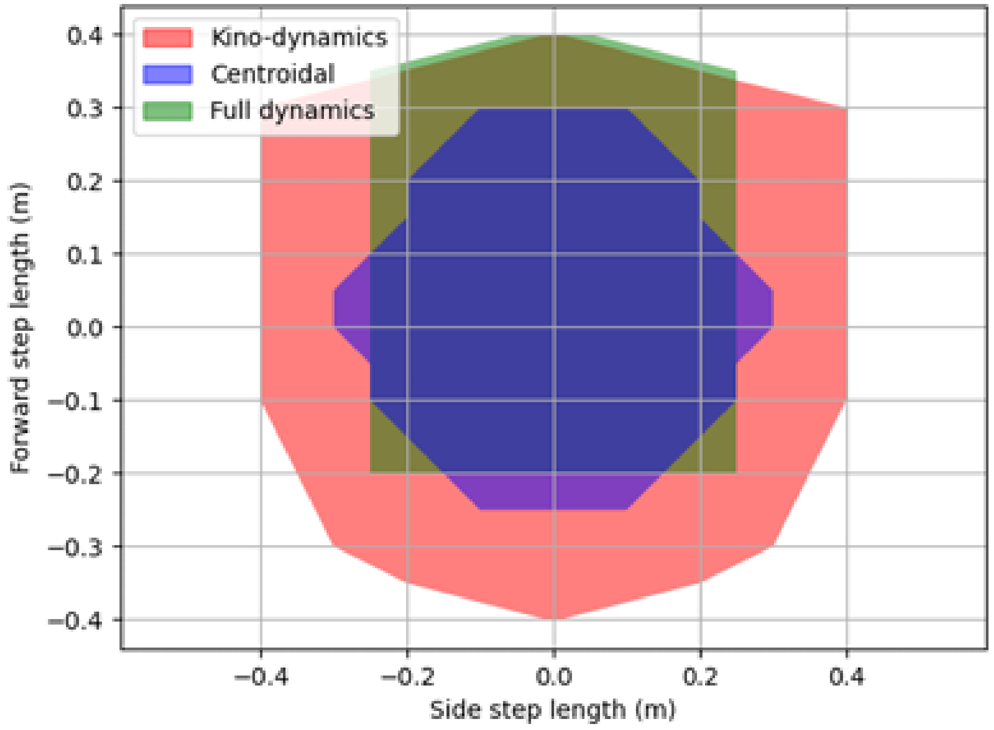
ATTENTION: Centroidal model is the only one that perform MPC in 10 ms

ASSUMPTION: every model works at 100 Hz

Model	Mean MPC Solving Time (ms)	Std Deviation (ms)	MPC Decision Variables	MPC Constraints
Centroidal	2.67	0.280	2.100	34
Kino-dynamics	20.3	1.62	9.100	45
Full dynamics	24.1	2.69	7.900	122



Simulation: Step Length



In this experiment, we measured the maximum step length that the proposed models are able to reach

The simulation is successful if the robot manages to take 4 steps before stopping

The kino-dynamic scheme is more robust than full dynamics and centroidal dynamics, which are quite comparable

SIMULATIONS

STRAIGHT TRAJECTORY

Simulation: Perturbations

In this simulation, the resilience of models was evaluated

A force of $0.1\ s$ duration is applied to the base frame from different angles with an intensity between 0 and $450\ N$

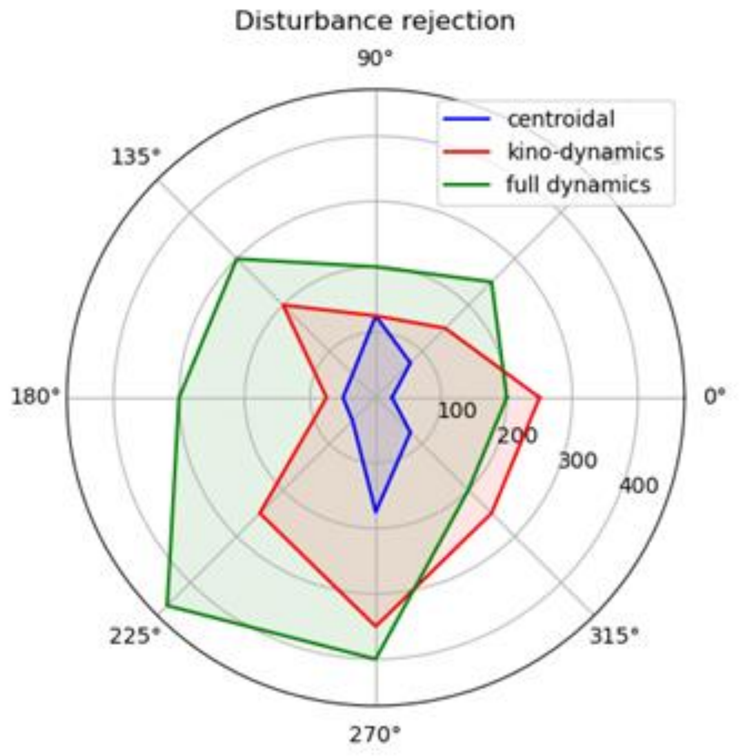
The robot follows a $20\ cm$ step and the force is applied when the right foot is at its highest point

Simulation: Perturbations

Both kino-dynamics and full-dynamics have better recovery capabilities than the centroidal model

Kino-dynamics is more robust to rear pushes

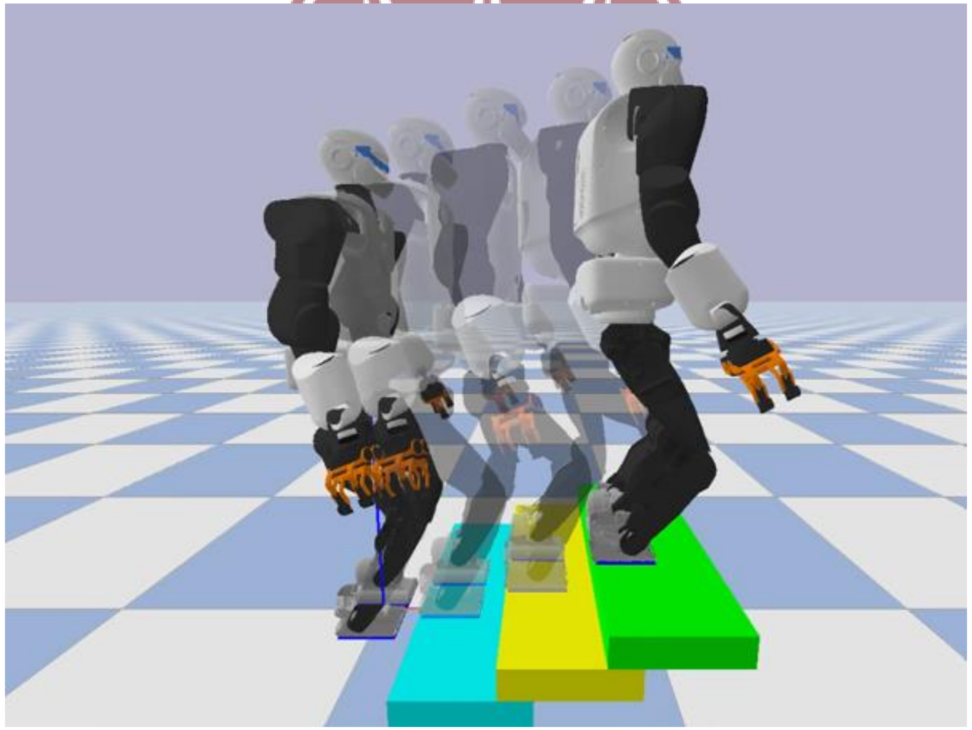
Full dynamics responds better to front pushes



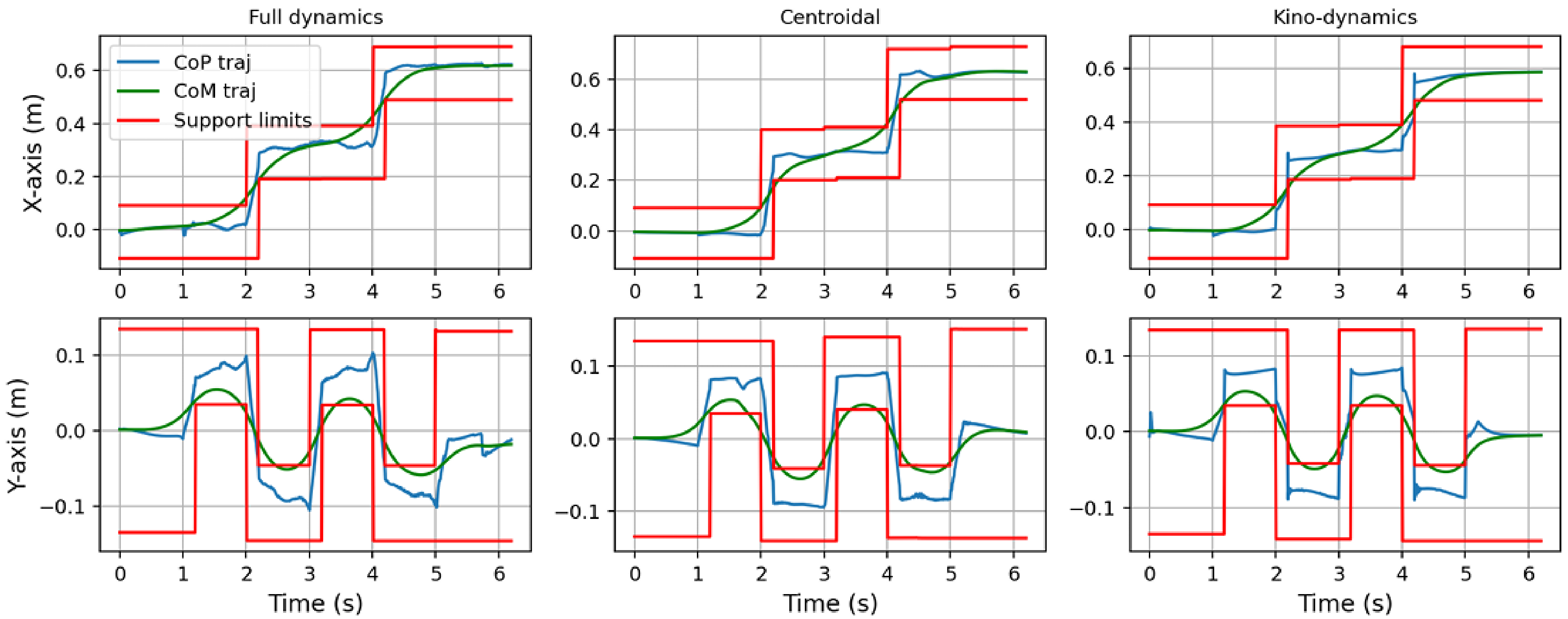
Push-offs to the right are handled better because they occur when the CoM is above the left foot, thus implying that the right foot can soon be used to counteract the applied force

Simulation: Stairs

In these simulations, the robot climbs a ladder, with the steps $10\ cm$ high



Simulation: Stairs



The image shows the trajectory of the CoP, CoM and support limits

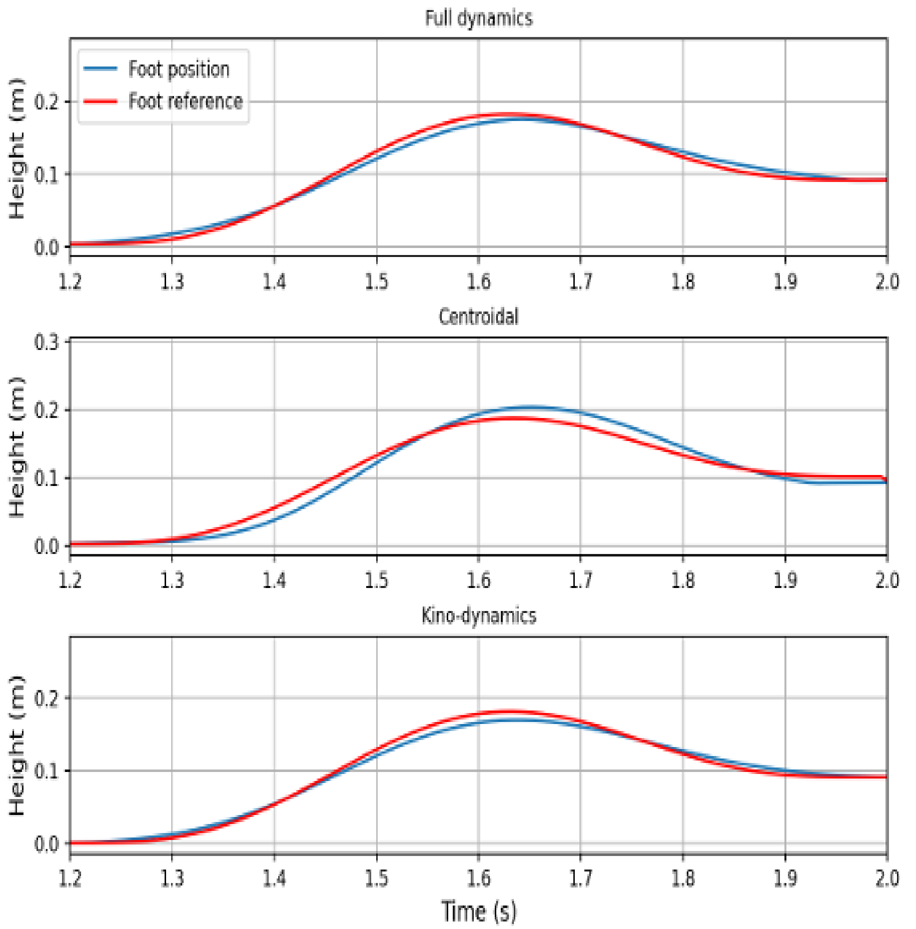
Note how kyno-dynamics has the smallest stability margins

Simulation: Stairs

The picture shows the vertical component of the left foot when climbing stairs

We can see that the centroidal model shows a delay in trajectory following

Because the centroidal model has no information on the future reference movement

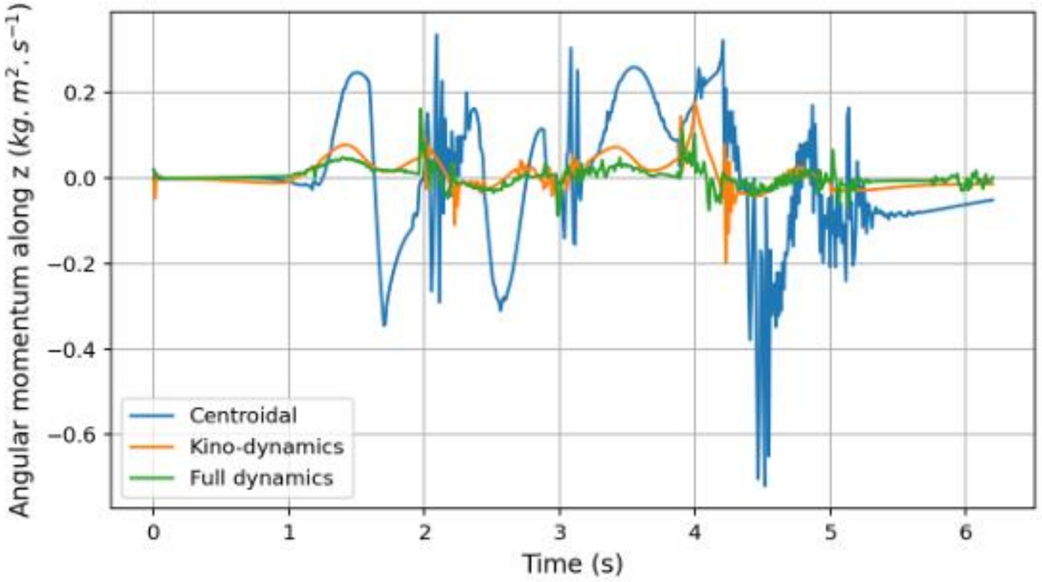


Simulation: Stairs

The picture shows the vertical component of the angular momentum

The centroidal model cannot have a small angular momentum

This is due to the fact that influence of whole body movement is only considered in the QP



Comparison Full-Body

Full-Body	
High precision: consider internal dynamic of the robot, including inertial effects of limbs	**Computational load:** takes longer to compute, limiting the frequency of re-planning
Feasibility: ensures kinematic and dynamic feasibility along the predictive horizon	**Complexity:** difficult to implement in real time on platforms with limited resources
Robustness: better management of disturbances	

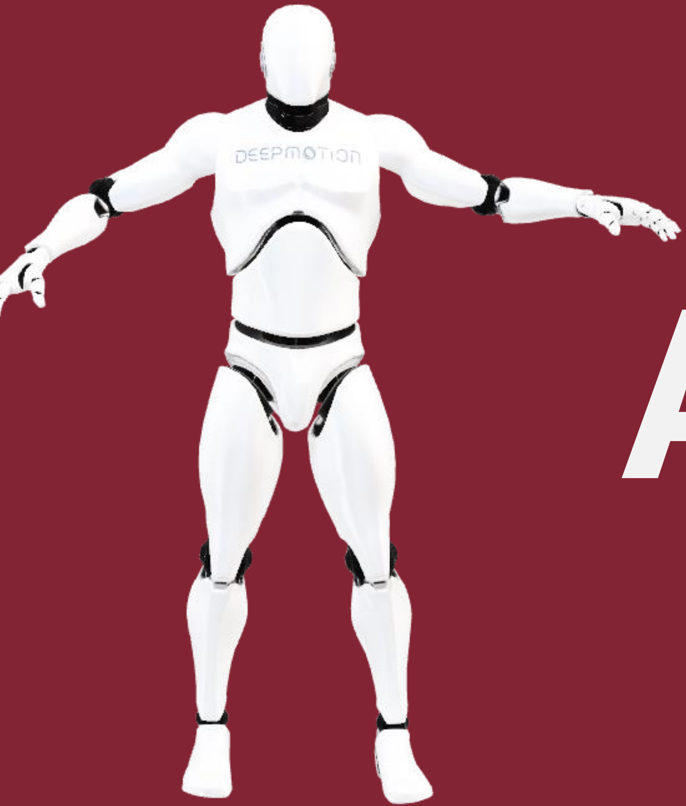
Comparison Centroidal

Centroidal	
Computational efficiency: very light and suitable for real-time applications	**Kinematic limitations:** does not guarantee kinematic and dynamic feasibility at joints
Low load: ideal for embedded systems with limited resources	**Reduced accuracy:** does not consider the inertial effects of limbs
	Limited robustness: less effective in recovering disturbances

Comparison Kino-dynamics

Kino-Dynamics	
Optimal compromise: combines centroidal dynamics with full robot kinematics	**Moderate computational load:** takes longer than the centroidal model, but less than full dynamics
Flexibility: allows kinematic tasks and constraints to be considered along the predictive horizon	**Dynamic limitations:** does not fully consider the inertial effects of limbs
Robustness: good ability to recover from pertubations	

THANKS FOR THE ATTENTION



THANKS FOR THE ATTENTION